

# puremacro: browser-runnable empirical macroeconomics in pure Python

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## Summary

**puremacro** is a Python library for empirical macroeconomics and quantitative macro modeling, implemented entirely on the scientific-Python core — NumPy (Harris et al. 2020), SciPy (Virtanen et al. 2020), pandas, and Matplotlib — with no compiled or specialized dependencies at runtime. Because it avoids heavier libraries, it executes unchanged in the browser through Pyodide (The Pyodide development team 2024), on a tablet, or in a JupyterLite notebook, as readily as on a workstation. The library spans the methods an applied macroeconomist reaches for: reduced-form and structural vector autoregressions (Sims 1980; Blanchard and Quah 1989), factor-augmented VARs (Bernanke et al. 2005), local projections (Jordà 2005), ARCH/GARCH and GARCH-MIDAS volatility models (Engle 1982; Bollerslev 1986), heteroskedasticity- and autocorrelation-consistent inference (Newey and West 1987), dynamic-panel and weak-instrument estimators (Anderson and Rubin 1949; Montiel Olea and Pflueger 2013), difference-in-differences and synthetic control, value-function iteration, Sequence-Space HANK general equilibrium (Auclert et al. 2021), mixed-frequency nowcasting (Giannone et al. 2008), integrated climate macroeconomics (Nordhaus 2018; Golosov et al. 2014), and text-based uncertainty and narrative burst indices (Baker et al. 2016).

## Statement of need

Macroeconometrics and computational macro are usually taught and practiced with tools that demand a substantial local installation — the statsmodels (Seabold and Perktold 2010), **arch**, and **linearmodels** stack — or commercial software, such as MATLAB for Dynare and most heterogeneous-agent toolkits. That barrier is invisible to well-resourced users and decisive for everyone else: students on tablets, instructors in low-bandwidth classrooms, and researchers without budgets or administrative rights. **puremacro** removes it. Because the library is pure scientific-Python, a reader can open a browser and reproduce

a structural VAR, a local-projection impulse response, an Aiyagari/HANK equilibrium, or a real-time GDP nowcast at zero cost, with nothing to install.

Breadth alone is not trustworthy, so `puremacro` ships its own evidence. A built-in **validation gallery** contains 73 cases that continuously check the library’s estimators against independent references — `statsmodels` (Seabold and Perktold 2010), `arch`, and `linearmodels` where a canonical implementation exists, and closed-form solutions, published numbers, or internal cross-method identities elsewhere — across thirteen subsystems (Figure 1). A companion **replication gallery** reproduces published results from heterogeneous-agent, SVAR, and climate-economy models. Both are wired into the CI workflows shipped with the package and re-run in a single line, `puremacro.validation.scorecard()`, including in the browser, so a user can verify the library on the same machine that runs it. No existing macro-econometrics package, to our knowledge, combines this breadth with browser portability and a continuously verified, self-contained correctness record.

## Features

- **Time series, SVARs, and Identification:** VAR, FAVAR, BVAR (Minnesota), VECM, and TVP-VAR; structural identification via Cholesky, long-run (Blanchard-Quah), sign and sign-zero restrictions, proxy/external instruments, and narrative signs; impulse responses, variance decompositions, and bootstrap bands.
- **Nowcasting and Machine Learning:** mixed-frequency Dynamic Factor Models (DFM) with ragged-edge EM imputation and news decomposition; Elastic Net and Adaptive Lasso penalized forecasting.
- **Single-equation and Panel:** local projections (including IV and panel), dynamic-panel GMM, HAC and weak-instrument-robust Anderson-Rubin inference, synthetic difference-in-differences, and synthetic control.
- **Computational and Climate Macro:** Sequence-Space HANK general equilibrium solve, value-function iteration, the endogenous grid method, continuous-time HJB, and DICE-2016 integrated climate-economy simulations.
- **Reproducibility and Teaching:** an in-browser JupyterLite playground with 36 bilingual (English/Spanish) interactive showcase notebooks and empirical datasets.

## Acknowledgements

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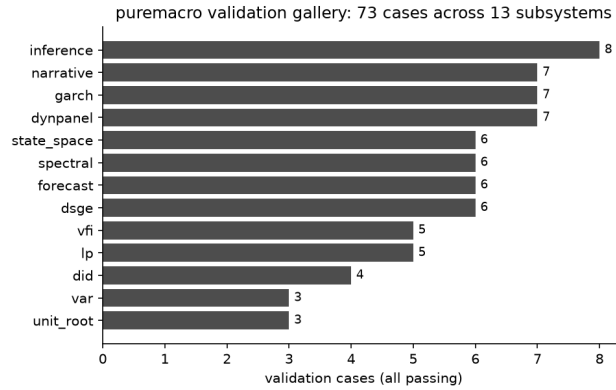


Figure 1: Validation coverage: 73 cases across 13 subsystems, each checked against an independent reference.

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